Predicting Spotify Hits: A Multimodal Classification Modeling Approach

— Inspired by HAIM's multimodal learning philosophy

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01 Motivation & Goal

- 100K+ new tracks uploaded daily; <1% succeed
- Success depends on artist influence + content quality
- Goal: build a model that predicts whether a song becomes a hit
- Approach: unify all song signals through multimodal modeling

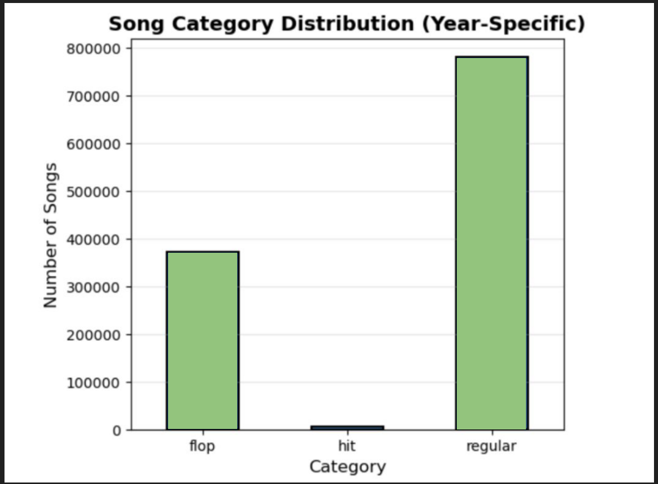
02 Data Overview

- Base: 1.16M Spotify tracks with audio features + metadata
  (Examples: danceability, energy, loudness, artist name...)

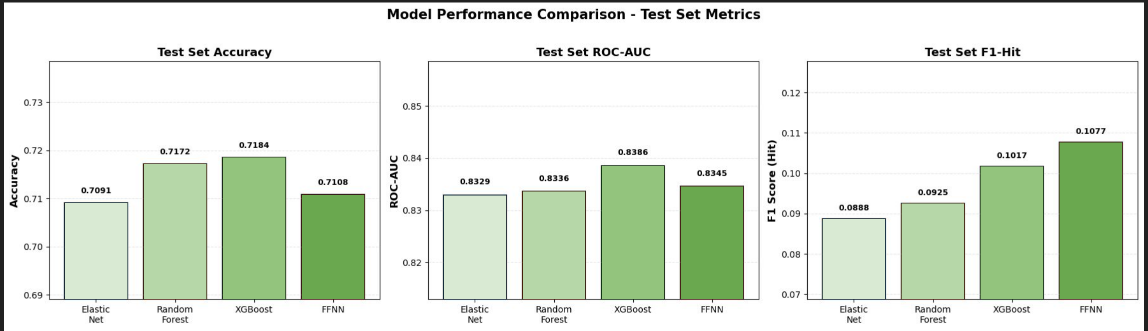
- Additional embedded modalities:
  - generated text description
    (Examples: "This song, titled ['track_name'], is performed by ['artist_name'], ...)
  - 30s audio previews
  - lyrics
  - album covers

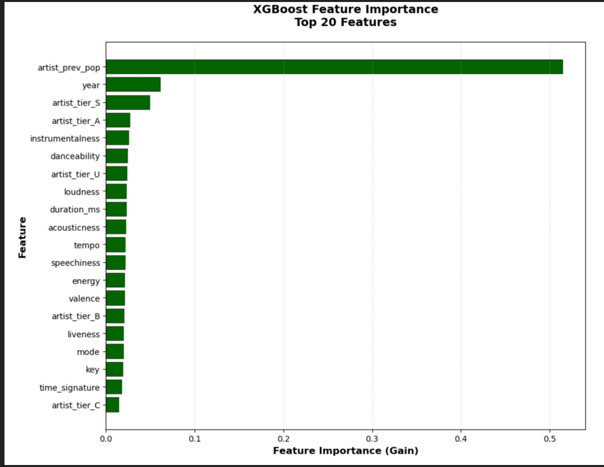
03 Key Preprocessing

- **Artist tier derived from Last.fm listener percentiles**
  (Last.fm listener percentile → S/A/B/C/F/U tiers)
- **artist_prev_pop = historical average popularity**
  (Average past Spotify popularity; strong predictor of future performance)
- **Undersample majority classes**
  (5.5K per class to address extreme class imbalance)
- **Time-based split**
  (Train: 2000–2021; Test: 2022–2023 to mimic real forecasting)
- **Year-normalized popularity → <u>flop / regular / hit</u>**
  (Fixes bias toward newer songs; labels based on within-year percentiles)



04 Basline Model/Data Selection

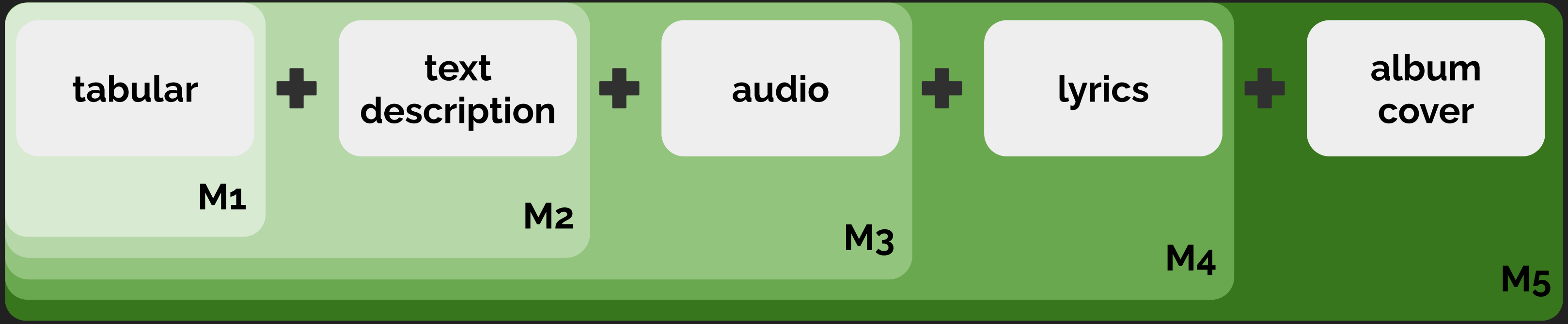


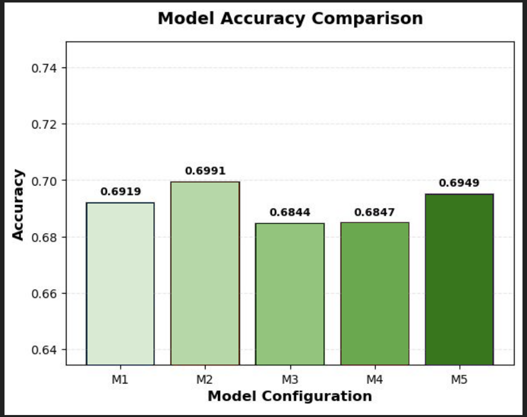


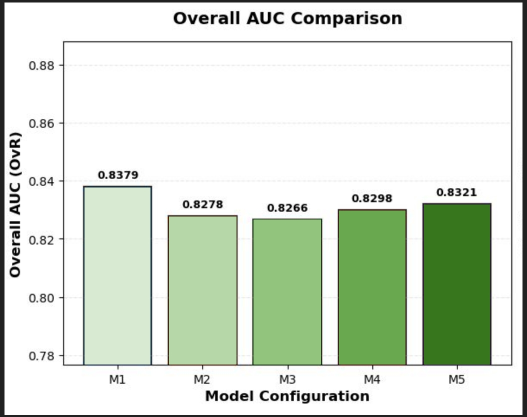
- XGBoost performs similarly to FFNN on tabular data
- FFNN handles high-dimensional multimodal embeddings better
- FFNN trains faster utilizing GPU
- → We select FFNN as the baseline.

- artist_prev_pop dominates all other features by a large margin
- Experiment models including and excluding this important feature

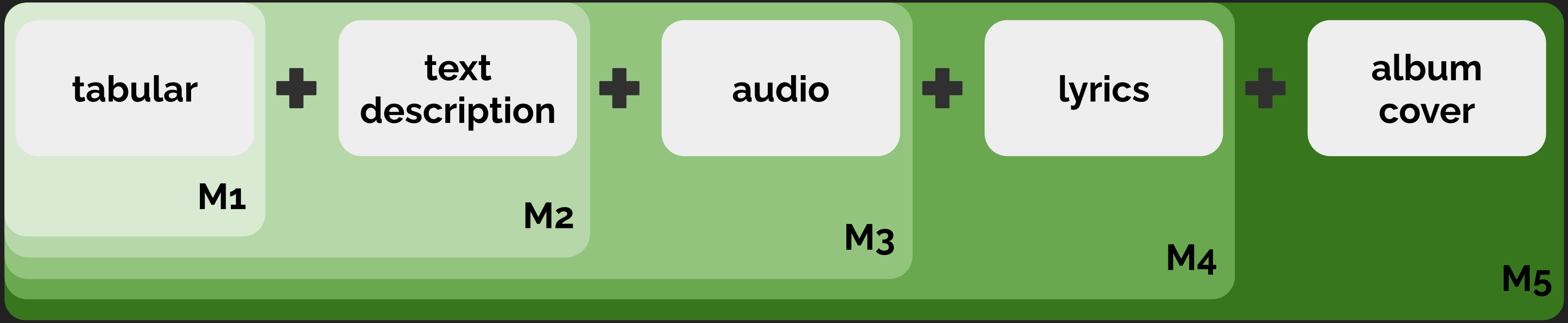
05 Multimodal Pipeline and Result: With artist_prev_pop

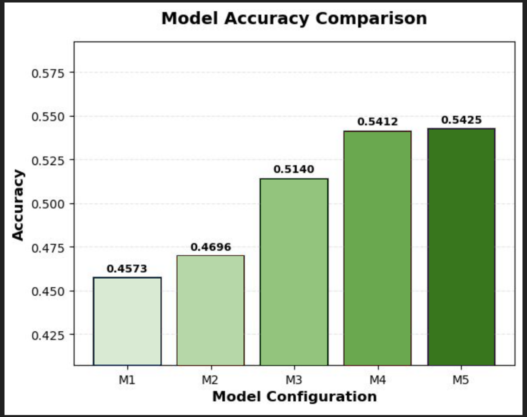


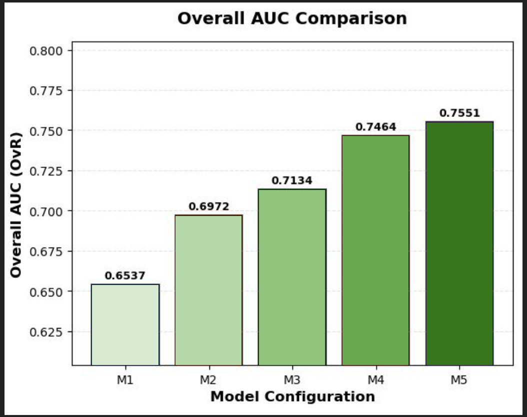




06 Multimodal Pipeline and Result: Without artist_prev_pop







07 Discussion and Conclusion

Key Insights:

- Artist history drives song success more than content
- Multimodal signals matter when history is weak or unavailable
- Best use cases: cold-start recommendations, new artists, early content screening

Future Steps:

- Train on full 1.16M tracks
- Explore XGBoost as baseline model
- Evaluate all modality combinations and identify which modes contribute the most
- Use longer audio segments (e.g., 30s → 2 minutes) for more musical context

Thank you!